

# SWARNA GOWRI MADAKALA

Gen AI Engineer | Front-End Developer | LangChain | Azure OpenAI | React | Python | Full-Stack AI Apps  
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## SUMMARY

Gen AI Engineer and Front-End Developer skilled in building production-grade AI applications using Python, LangChain, Azure OpenAI, and React.js. Experienced in RAG pipelines, prompt engineering, agentic workflows, and full-stack deployment across AWS, GCP, and Azure. Proficient in integrating OpenAI GPT-4, Google Gemini, and Hugging Face into scalable web apps with measurable improvements in accuracy and performance.

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript (ES6+), HTML5, CSS3

**Frontend:** React.js, Responsive Web Design, DOM Manipulation, Component Architecture

**Gen AI & LLM:** Azure OpenAI, OpenAI API, Hugging Face, Google Gemini, LangChain, RAG, Prompt Engineering, Guardrails, Vector Search, Embeddings

**AI Frameworks:** LangChain, TensorFlow, PyTorch, scikit-learn

**Cloud & Deployment:** AWS, GCP, Azure (AI Foundry, AI Search), REST API Integration, Cloud App Deployment

**Agentic AI:** Agentic Workflows, Role-Based Retrieval, Confidence Heuristics, Fallback Flows, Eval Pipelines

**NLP & ML:** Text Preprocessing, TF-IDF, Word Embeddings, Sentiment Analysis, Classification, Regression

**Data & Tools:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Git, GitHub, VS Code, Jupyter Notebook

## PROJECTS

### AI-Powered Learning Assistant — React Web Application

*React.js | OpenAI API | Google Gemini | Hugging Face | REST API | Prompt Engineering | GCP | CSS3 | HTML5*

- Built a full-stack AI-powered learning platform with React.js frontend integrating OpenAI GPT-4 and Google Gemini APIs to generate personalized course content and quizzes dynamically.
- Engineered multi-model prompt workflows — switching between OpenAI and Gemini based on task type (content generation vs. Q&A reasoning), improving response relevance by ~35%.
- Integrated Hugging Face sentiment analysis API to analyze learner feedback in real-time and adapt content difficulty accordingly.
- Deployed on GCP (Google Cloud Run) with CI/CD pipeline — enabling zero-downtime updates for 500+ simulated users.
- Designed fully responsive UI with CSS Grid and Flexbox — optimized for mobile, tablet, and desktop across all major browsers.

### GenAI Patient Support Knowledge Assistant — Full-Stack RAG Application

*Python | Azure OpenAI | LangChain | Azure AI Search | Guardrails | REST API | JSON Outputs | Client: HealthPlus Hospitals (MEA)*

- Architected end-to-end RAG pipeline with role-based retrieval controls, metadata filters, and structured JSON API outputs consumed directly by a React-based support UI.
- Engineered safe prompts with medical-advice guardrails, automatic disclaimers, and escalation routing to human agents — production deployed on Azure Cloud.
- Built confidence scoring (retrieval score thresholds) with graceful fallback flows — system handles edge cases without hallucinating responses.
- Delivered eval pipelines tracking accuracy and refusal correctness across iterations — drove measurable quality improvements in production.
- Integrated LangChain agent workflows for dynamic tool selection between FAQ retrieval, live escalation, and structured data lookup.

### AI Content Generation Tool — Prompt Engineering & Google Gemini Integration

*Python | Google Gemini API | OpenAI API | Hugging Face Transformers | LangChain | REST APIs | HTML/CSS | AWS*

- Built a web-based AI content studio using Google Gemini and OpenAI APIs to generate SEO-friendly blog posts, social media copy, and course descriptions from structured prompts.
- Designed a layered prompt engineering framework — system prompts, few-shot examples, and chain-of-thought instructions — reducing output hallucinations by 40%.
- Integrated Hugging Face Transformers for text summarization and keyword extraction as preprocessing steps before sending to Gemini for generation.
- Deployed on AWS (EC2 + S3) with a lightweight HTML/CSS/JavaScript frontend — accessible via public URL with API key-based authentication.

### Loan Default Risk Prediction — Machine Learning

*Python | scikit-learn | TensorFlow | REST API | Machine Learning*

- Built binary classification model on demographic, financial, and transactional features to predict loan default risk using Decision Trees, Random Forest, and Logistic Regression.
- Implemented a TensorFlow neural network baseline for comparison — evaluated using ROC-AUC, cross-validation, and Platt Scaling for calibrated probability outputs.
- Exposed the trained model as a REST API endpoint — enabling integration with third-party business decision frameworks and web dashboards.

## INTERNSHIP EXPERIENCE

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### Data Science Intern | AI ML Labs Pvt. Ltd., Bengaluru

*6 Months | 2025*

- Ran EDA on structured datasets — surfaced feature correlations and outliers that directly shaped model design decisions.
- Built and tuned classification and regression pipelines: encoding, scaling, imputation, and cross-validation from scratch.
- Automated preprocessing workflows with reusable Pandas and NumPy notebooks, cutting experiment setup time across projects.
- Translated model performance metrics into Matplotlib and Seaborn dashboards for non-technical business stakeholders.

## CERTIFICATIONS

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**Java Programming Course** — Coding Ninjas | 2024

**Google Developer Student Club (GDSC) Certification** — Google | 2024

## EDUCATION

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### B.Tech in Computer Science & Engineering

*2022 – 2026*

Sri Sai Institute of Technology and Science, Rayachoty, Andhra Pradesh